

DHRUV VERMA

Ph.D. Candidate | Computational Imaging, Sensing, and Perception
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EDUCATION

University of Toronto

Ph.D. in Computer Science | Advisor: Alex Mariakakis
Expected Graduation: Fall 2026

Toronto, Canada
2021 - Present

Indraprastha Institute of Information Technology (IIIT) Delhi

B.Tech. in Computer Science & Engineering

Delhi, India
2017 - 2021

RESEARCH EXPERIENCE

Research Intern | Samsung AI Centre Toronto

- Coded-exposure imaging for power-efficient visual perception.

Apr 2026 - Present

Graduate Student | University of Toronto

- High-speed hyperspectral imaging and video of dynamic scenes using programmable light sources, smartphone cameras, and coded-exposure image sensors.
- Designed end-to-end imaging pipelines spanning hardware prototyping, sensor and illumination characterization (noise, spectral sensitivity, spectral power distribution, timing), image formation modeling, and reconstruction using optimization and deep learning.
- Developed applications in material-aware perception, fluorescence microscopy, and spectral lighting for color perception augmentation.

Sep 2021 - Present

Research Intern | Microsoft Research

- Developed novel smartphone-based sensing technique for non-invasive screening of Glaucoma.
- Built acoustic actuation and video-based Eulerian motion sensing pipelines to measure subtle eye surface deformation.
- Designed and validated prototype hardware and sensing algorithms using ex vivo porcine eye models.

May 2024 - Jul 2024

Applied Scientist Intern | Amazon Science

- Developed self-supervised contrastive learning models for hand

Jul 2022 - Oct 2022

gesture recognition from wearable IMU and PPG signals.

- Improved recognition accuracy by 12% while reducing reliance on labeled training data.

Visiting Researcher | Carnegie Mellon University

Jun 2020 – May 2021

- Developed multimodal machine learning technique for human activity recognition using wearable IMUs and low-resolution audio.
- Conducted wearable data collection, signal processing, multimodal fusion, and ablation studies for robust activity recognition.

Student Researcher | IIIT Delhi

Dec 2018 - May 2021

- Developed a novel sensing approach for facial expression recognition using IMU signals from wireless earbuds, enabling hands-free micro-gesture input and emotion detection.
- Built deep learning pipelines for detecting emotion and attention states from wearable EEG data in classroom settings.

PUBLICATIONS

A Design Space for Space-Time Manipulations Using Coded-Exposure Cameras

Under Submission

Lumosaic: Hyperspectral Video via Active Illumination and Coded-Exposure Pixels

Dhruv Verma, Andrew Qiu, Roberto Rangel, Ayandev Barman, Hao Yang, Chenjia Hu, Fengqi Zhang, Roman Genov, David B. Lindell, Kiriakos N. Kutulakos, Alex Mariakakis

CVPR 2026

ChromaFlash: Snapshot Hyperspectral Imaging Using Rolling Shutter Cameras

Dhruv Verma, Ian Ruffolo, David B. Lindell, Kiriakos N. Kutulakos, and Alex Mariakakis

IMWUT 2024. <https://doi.org/10.1145/3678582>

AttentioNet: Monitoring Student Attention Type in Learning with EEG-Based Measurement System

Dhruv Verma, Sejal Bhalla, Sai Santosh, Saumya Yadav, Aman Parnami, and Jainendra Shukla

ACII 2023. <https://doi.org/10.1109/ACII59096.2023.10388212>

SAMoSA: Sensing Activities with Motion and Sub-sampled Audio

Vimal Mollyn, Karan Ahuja, Dhruv Verma, Chris Harrison, and Mayank Goel

IMWUT 2022. <https://doi.org/10.1145/3550284>

ExpressEar: Sensing Fine-Grained Facial Expressions with Earables

Dhruv Verma, Sejal Bhalla, Dhruv Sahnan, Jainendra Shukla, and Aman Parnami

IMWUT 2021. <https://doi.org/10.1145/3478085>

Fashionist: Personalizing Outfit Recommendation for Cold-Start Scenarios

Dhruv Verma, Kshitij Gulati, Vasu Goel, and Rajiv Ratn Shah

Multimedia 2020. <https://doi.org/10.1145/3394171.3414446>

Addressing the Cold-Start Problem in Outfit Recommendation Using Visual Preference Modelling

Dhruv Verma, Kshitij Gulati, and Rajiv Ratn Shah

BigMM 2020. <https://doi.org/10.1109/BigMM50055.2020.00043>

AWARDS & HONORS

Special Recognition for Outstanding Reviews, IMWUT	2025
Beatrice "Trixie" Worsley Graduate Scholarship (\$4,000)	2025
University of Toronto Program-Level Fellowship (4× \$1,000)	2021 - 2024
ACM SIGCHI Gary Marsden Award (\$2,500)	2023
University of Toronto Departmental Fellowship (\$1,355)	2023
Alberta Machine Intelligence Institute AI Week Talent Bursary (\$1,500)	2022
Mitacs Globalink Research Internship Award (\$6,000)	2020

INVITED TALKS

Microsoft Research, "Seeing Beyond" with Everyday Cameras	2024
University of Toronto, Towards Practical Hyperspectral Imaging	2024
Amazon Science, Hand Gesture Recognition on Wrist-Worn Devices	2022

ACADEMIC SERVICE

Lead Editor, XRDS (<u>Computational Cameras</u>)	2024
Lead Editor, XRDS (<u>Augmented Humans</u>)	2022
Feature Editor, XRDS	2021 - 2024
Peer Reviewer, IMWUT, CHI, UIST, and MobileHCI	2021 - Present
Student Volunteer, Ubicomp and UIST	2020

TEACHING EXPERIENCE

Physical Computing and Interactive Technologies, Graduate Teaching Assistant	2025
Design of Interactive Computational Media, Graduate Teaching Assistant	2022 - 2025
AI for Medicine, Graduate Teaching Assistant	2022 - 2024
Introduction to Image Understanding, Graduate Teaching Assistant	2021
Prototyping Interactive Systems, Undergraduate Teaching Assistant	2019

STUDENTS ADVISED

Richard Yin, Undergraduate, Computer Science	2025 - Present
Tim Wang, Undergraduate, Computer Science	2025 - Present
Tanaya Sharma, Undergraduate, Computer Engineering	2025

Harpuneet Singh , Undergraduate, Computer Science	2024 - 2026
Andrew Qiu , Undergraduate, Computer Science (→ ML, Pinterest)	2024 - 2026
Yifang Zhang , Undergraduate, Computer Science	2024 - 2025
Shivesh Prakash , Undergraduate, Computer Science (→ Masters, UC Berkeley)	2022
Yifang Qu , Undergraduate, Computer Science (→ Masters, UToronto)	2019

VOLUNTEERING & OUTREACH

Social Coordinator , Dynamic Graphics Project Lab	2023
Mentor , University of Toronto <u>Pursue STEM Outreach Program</u>	2022